

2.3. Result of the Program and Analysis

[intro]

*Initial GUI

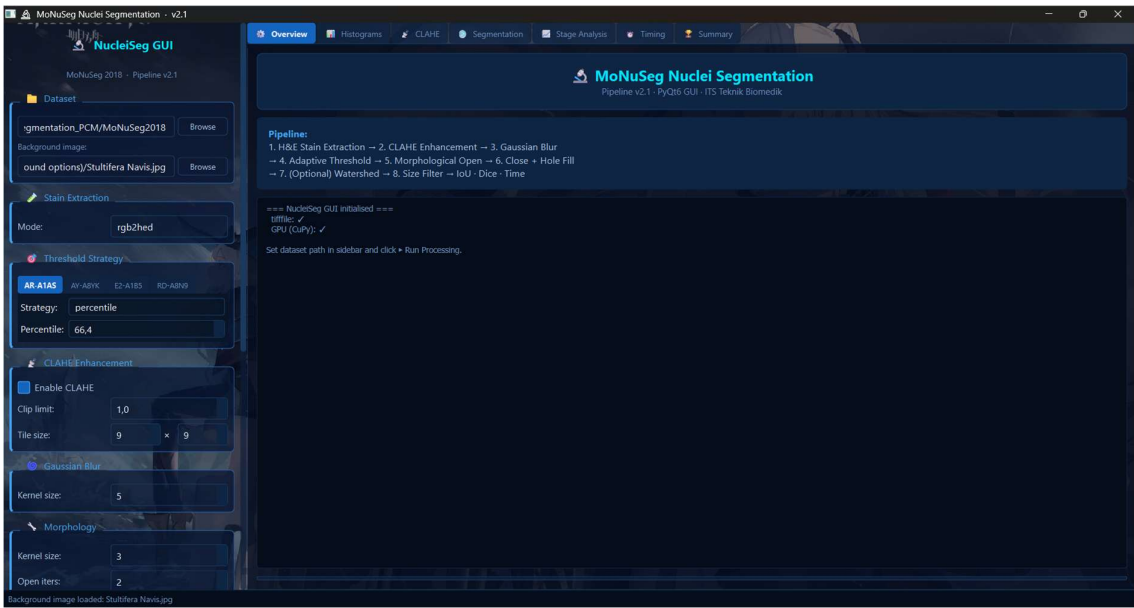


Figure 2.1. Initial GUI Window for the Nuclei Segmentation Program

[GUI explanation]

2.3.1. Parameter Initialization

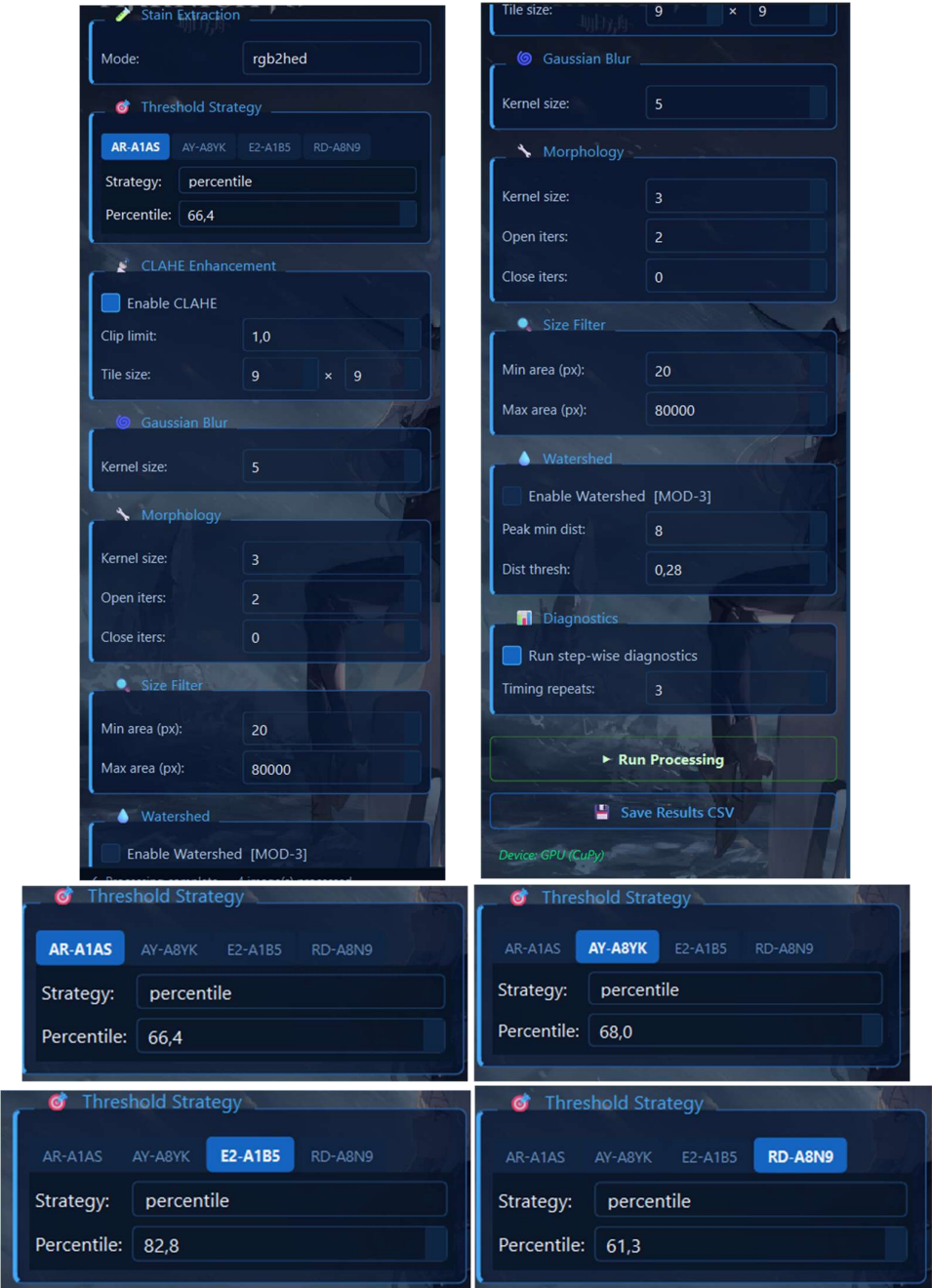


Figure 2.2. Initial Parameter Configuration (also Threshold Strategy for All Four TIF Image)

2.3.2. Histogram Results

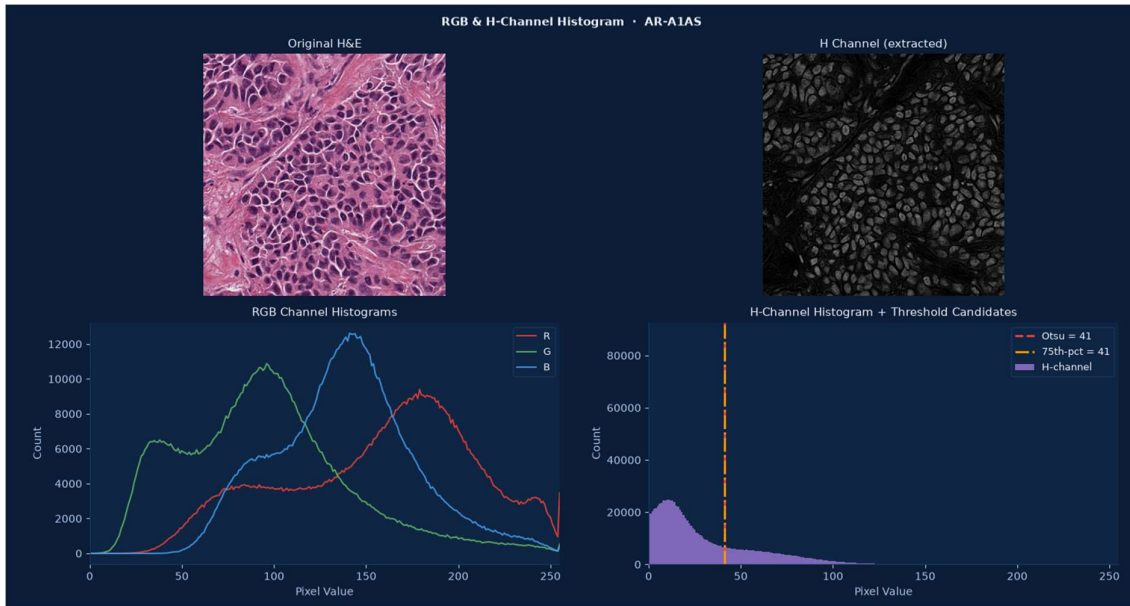


Figure 2.3. RGB and H-Channel Histogram for AR-A1AS Image

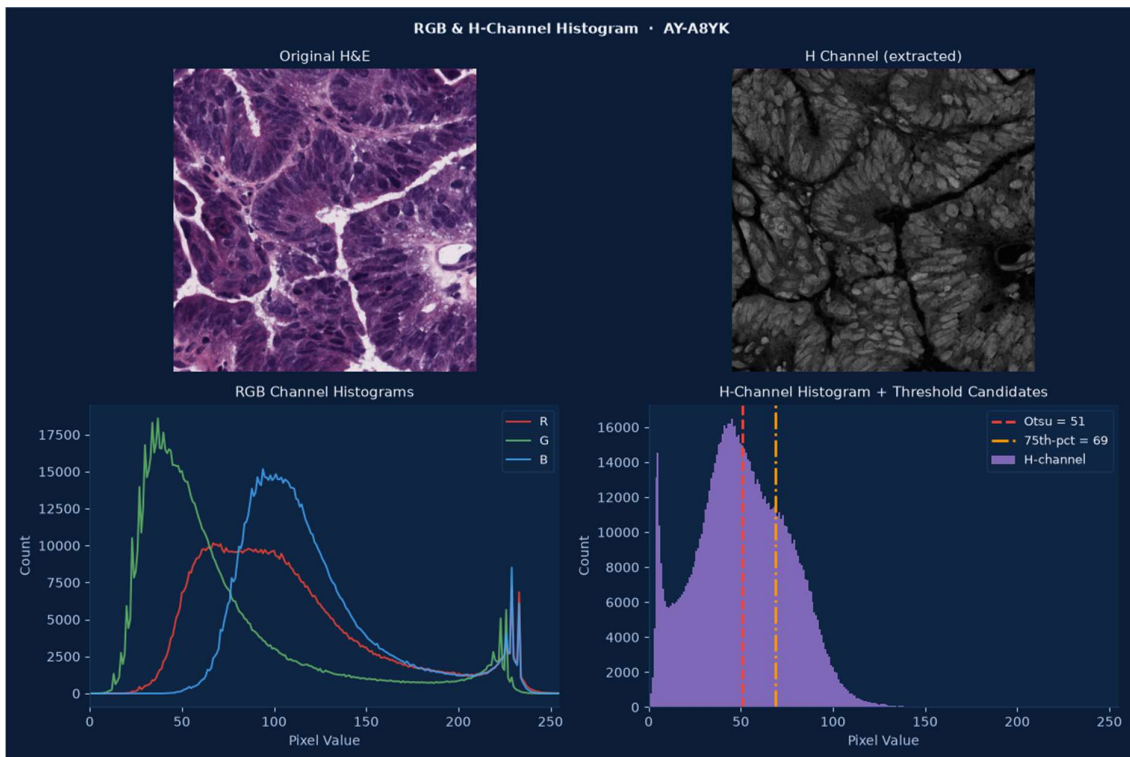


Figure 2.4. RGB and H-Channel Histogram for AY-A8YK Image

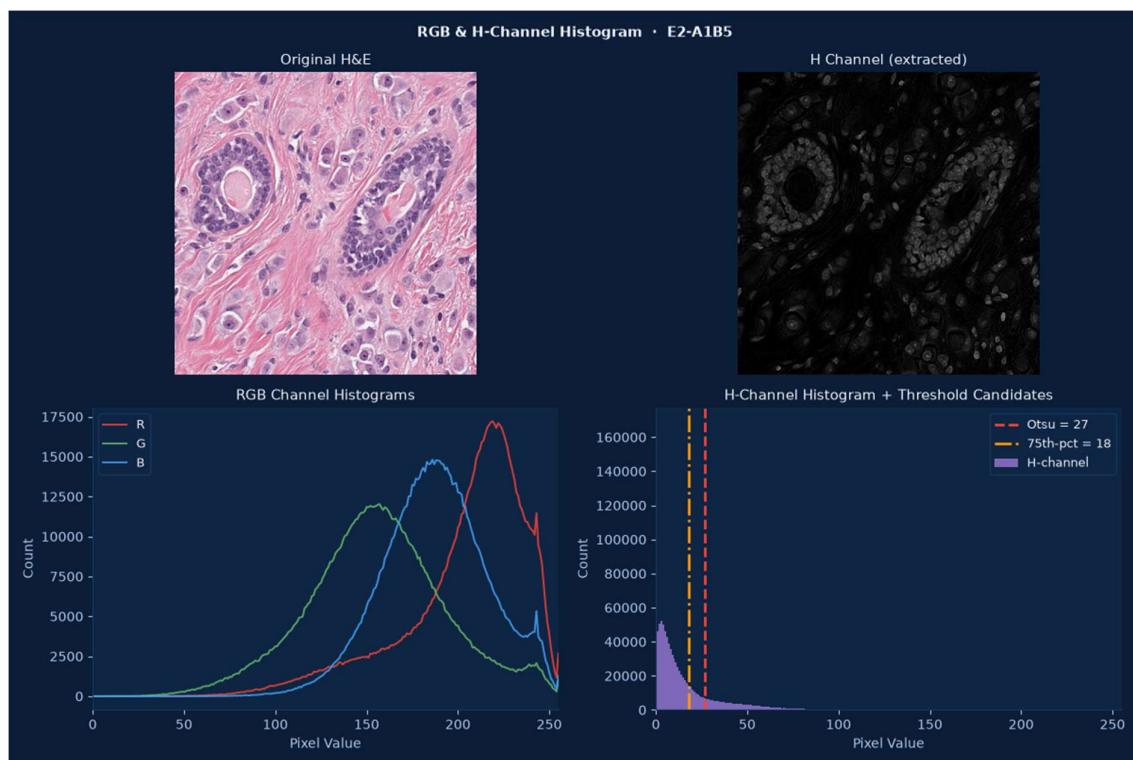


Figure 2.5. RGB and H-Channel Histogram for E2-A1B5 Image

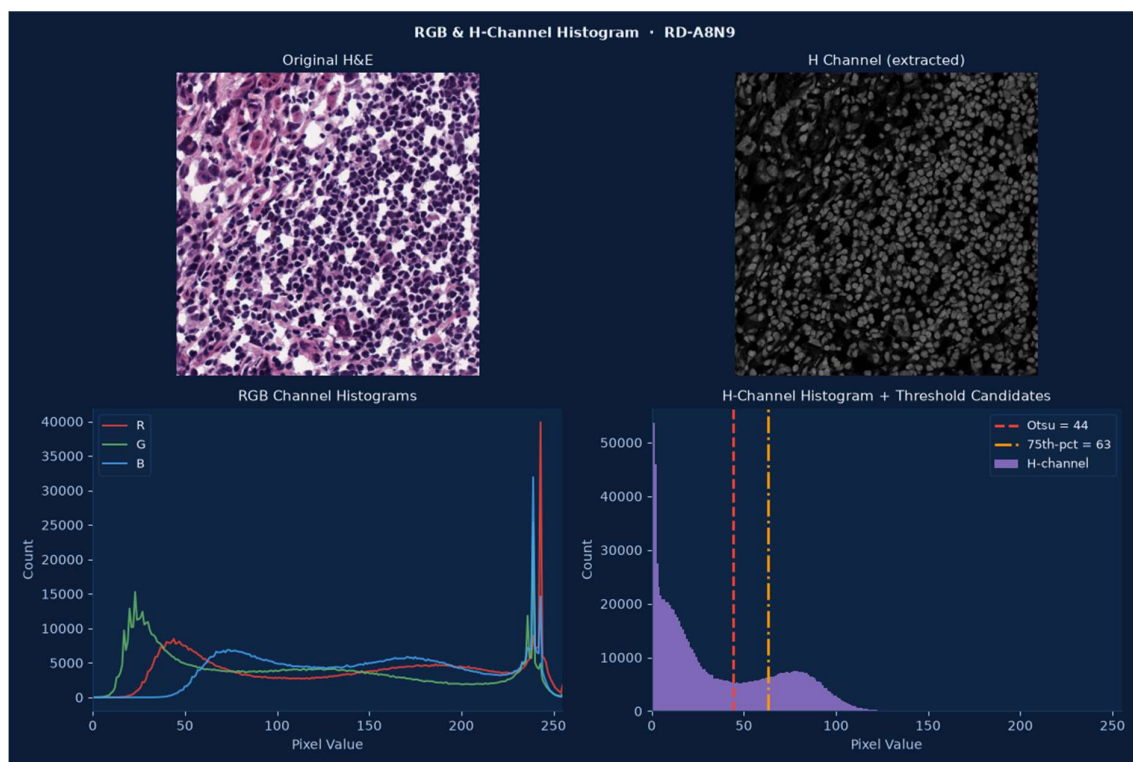


Figure 2.6. RGB and H-Channel Histogram for RD-A8N9 Image

2.3.3. CLAHE Results

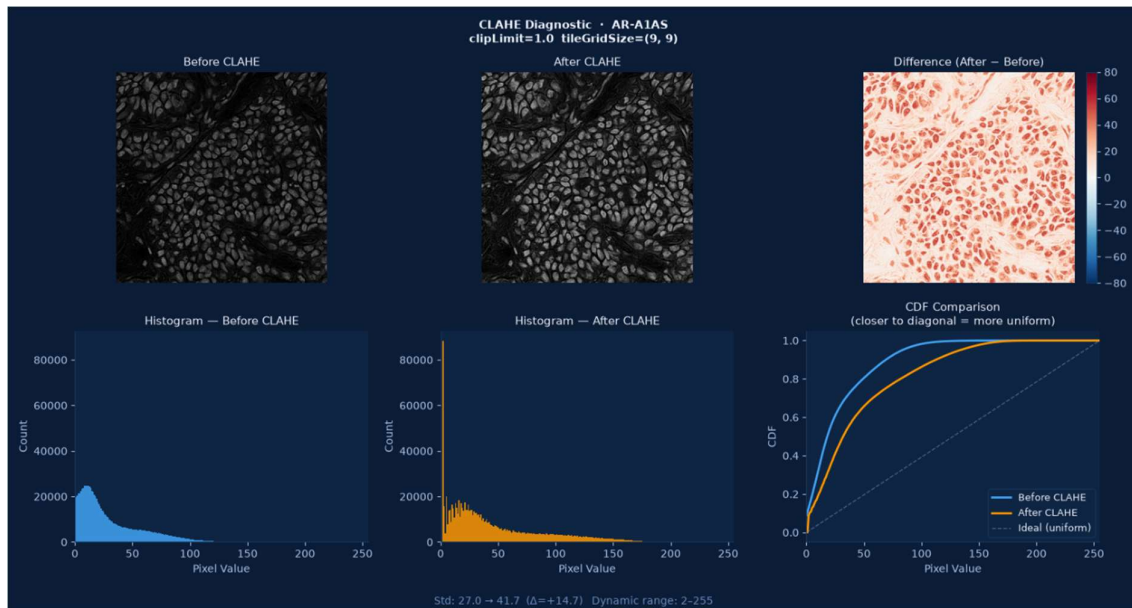


Figure 2.7. CLAHE Diagnostic for AR-AIAS Image with clipLimit=1.0 and tileGridSize=(9,9)

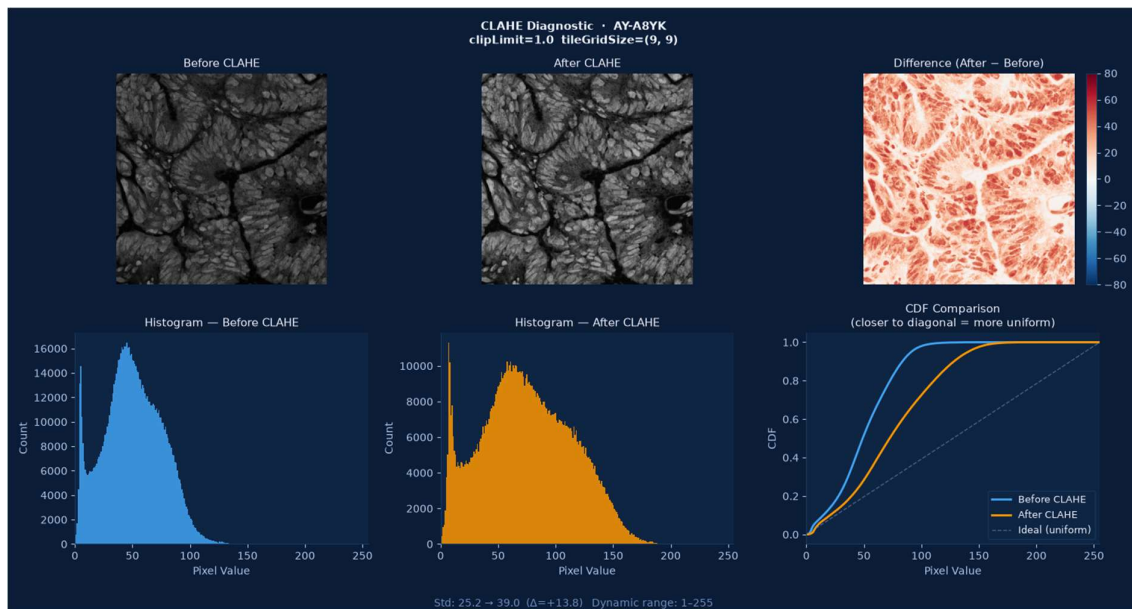


Figure 2.8. CLAHE Diagnostic for AY-A8YK Image with clipLimit=1.0 and tileGridSize=(9,9)

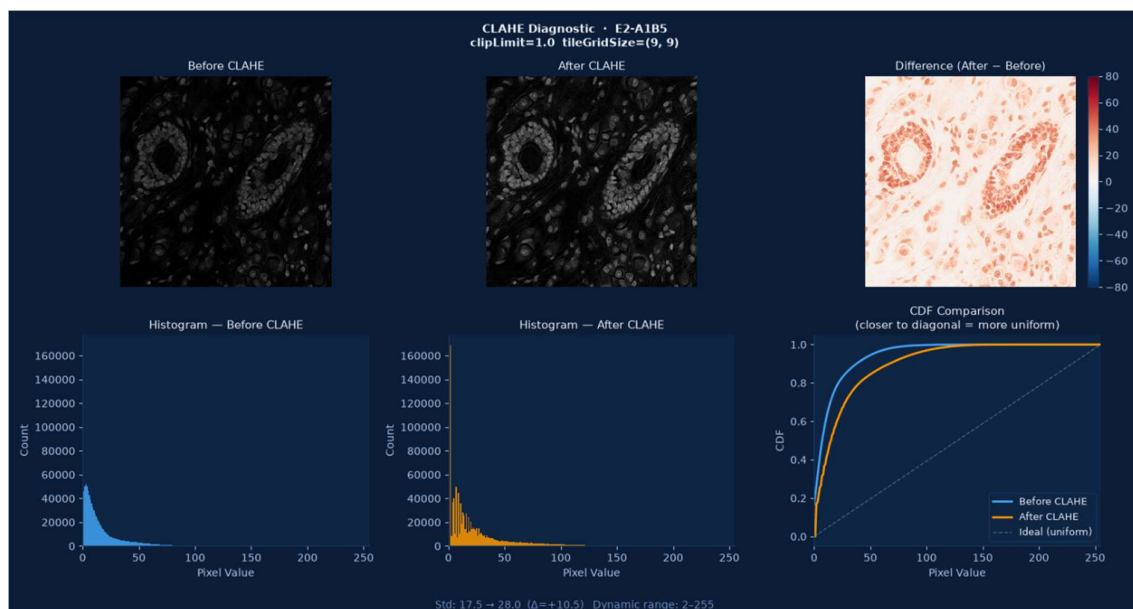


Figure 2.9. CLAHE Diagnostic for E2-A1B5 Image with clipLimit=1.0 and tileGridSize=(9,9)

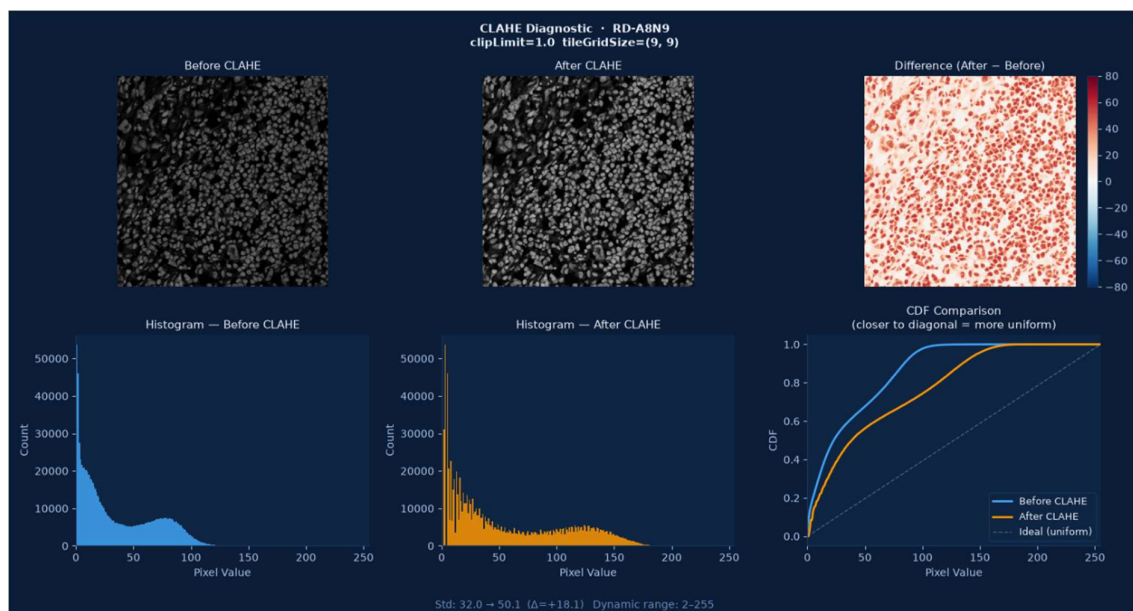


Figure 2.10. CLAHE Diagnostic for RD-A8N9 Image with clipLimit=1.0 and tileGridSize=(9,9)

2.3.4. Segmentation Results

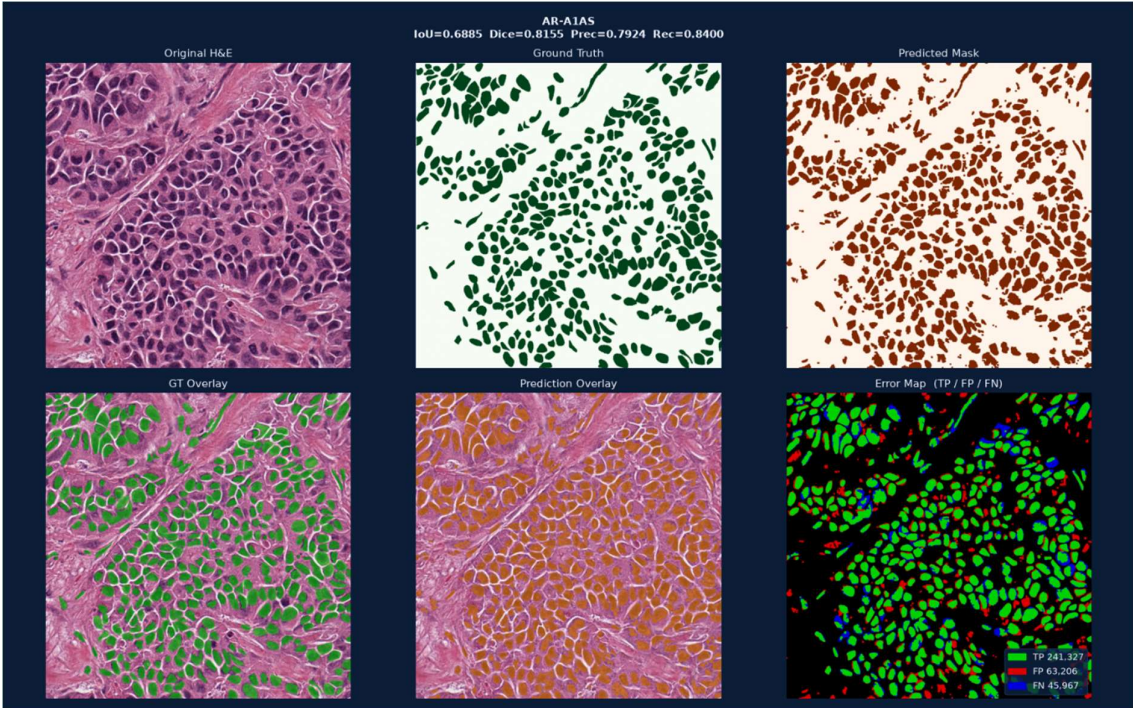


Figure 2.11. Segmentation Results for AR-AIAS Image

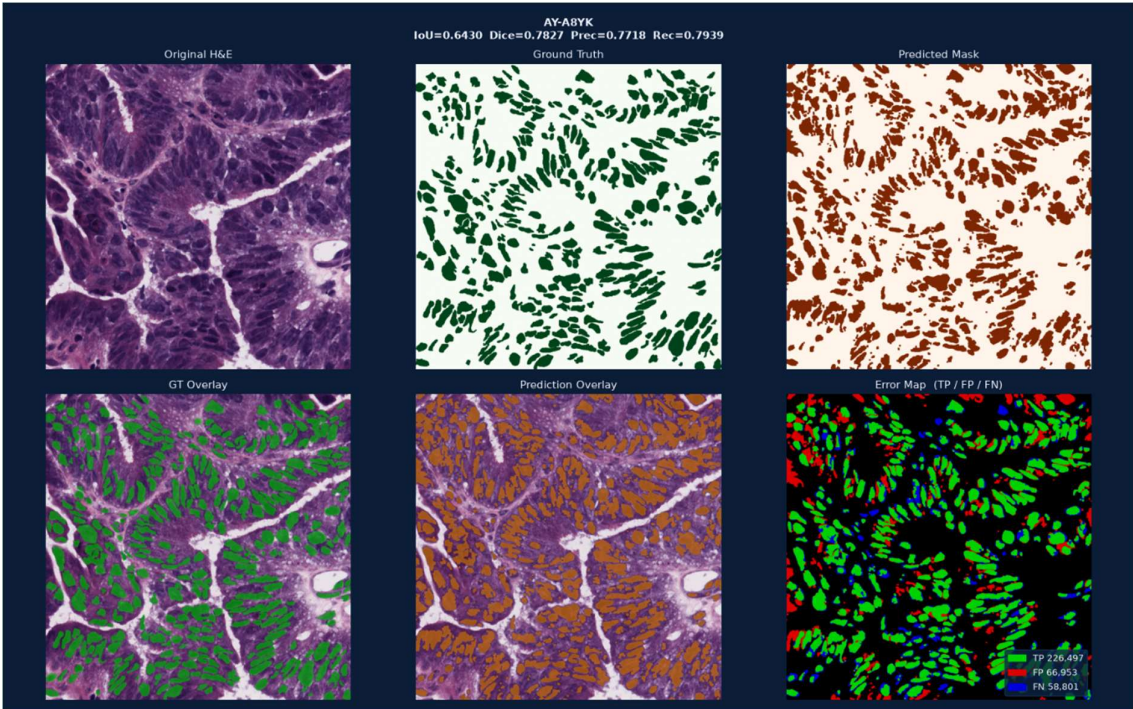


Figure 2.12. Segmentation Results for AY-A8YK Image

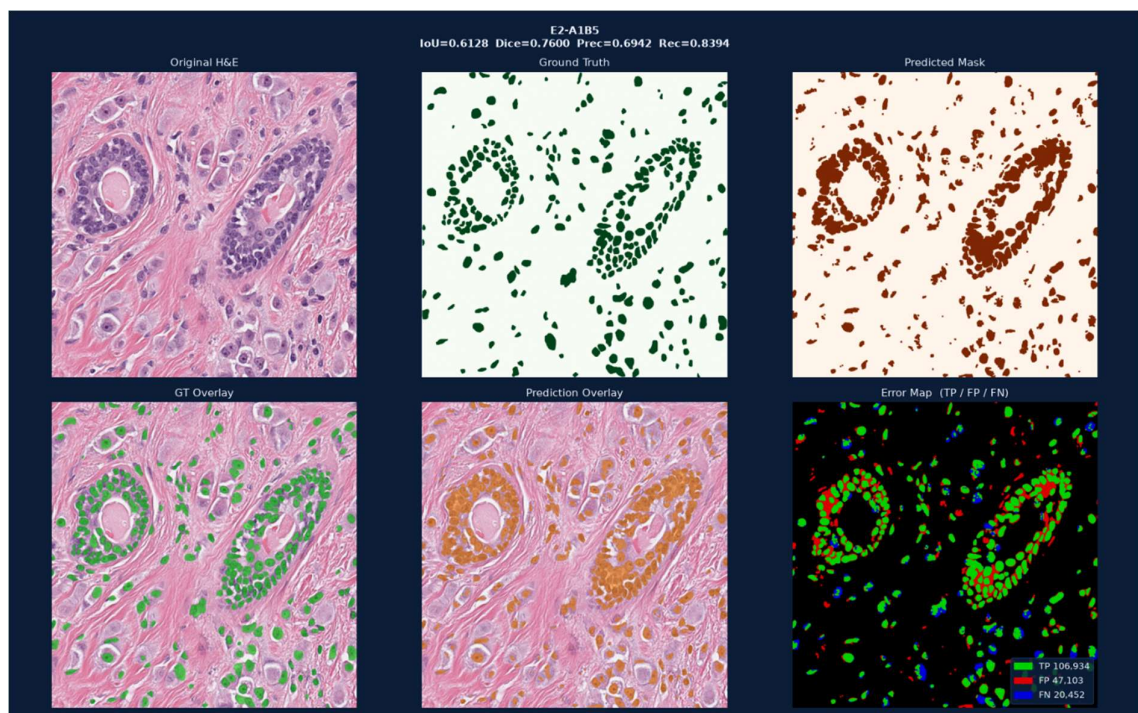


Figure 2.13. Segmentation Results for E2-A1B5 Image

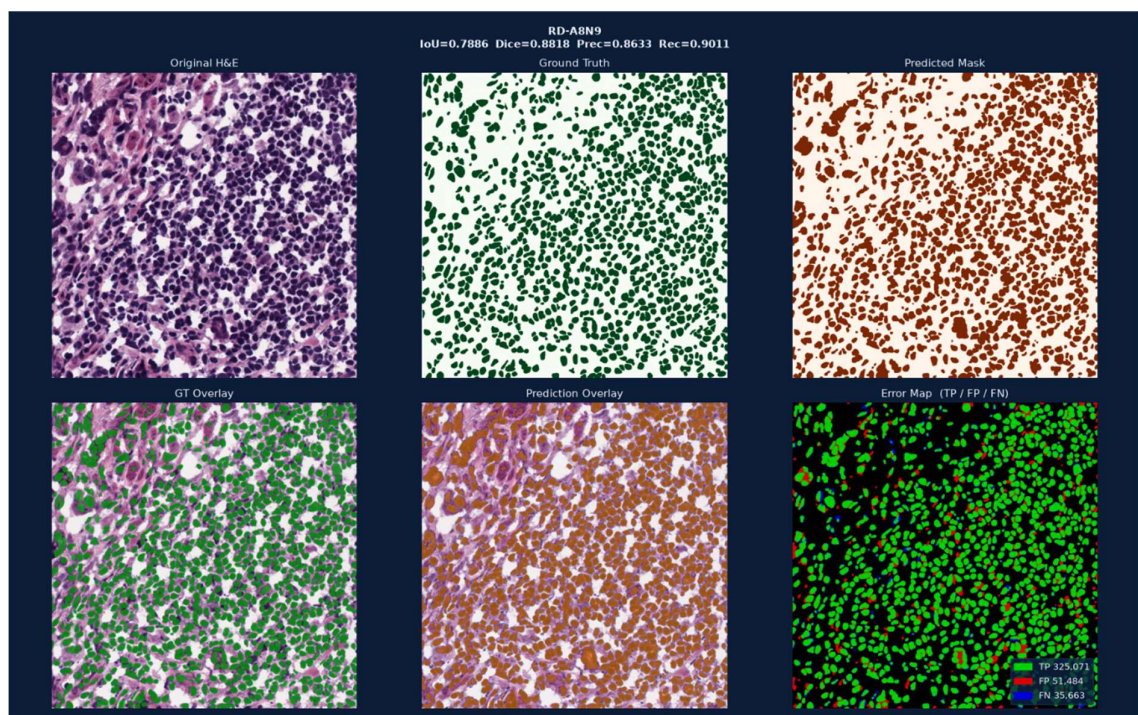


Figure 2.14. Segmentation Results for RD-A8N9 Image

2.3.5. Stage Analysis Results

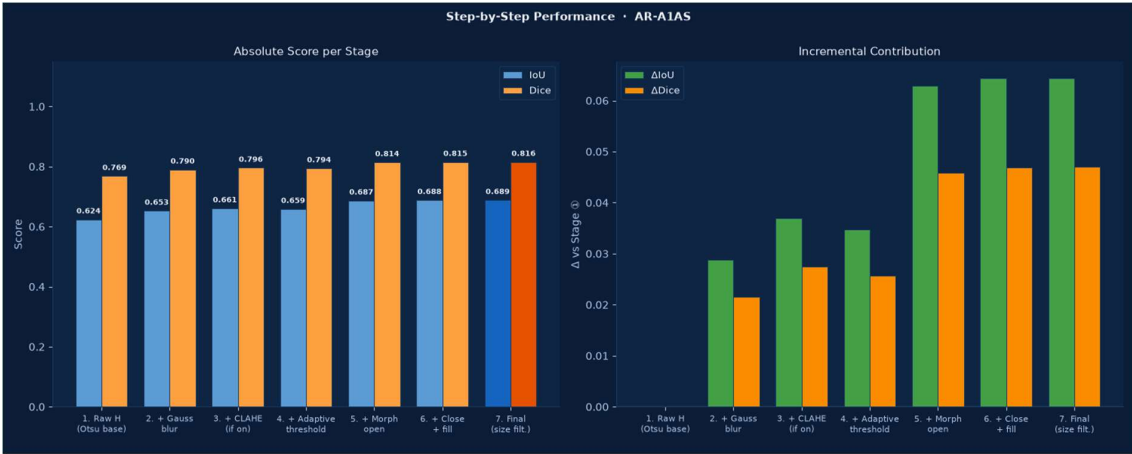


Figure 2.15. Step-by-Step Performance for AR-A1AS Image

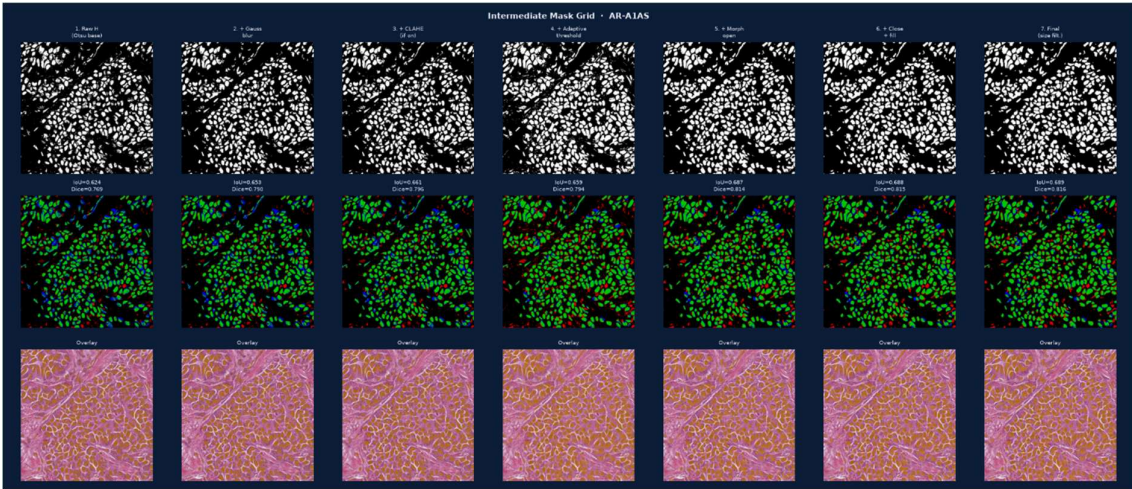


Figure 2.16. Intermediate Mask Grid Result on each Stage for AR-A1AS Image

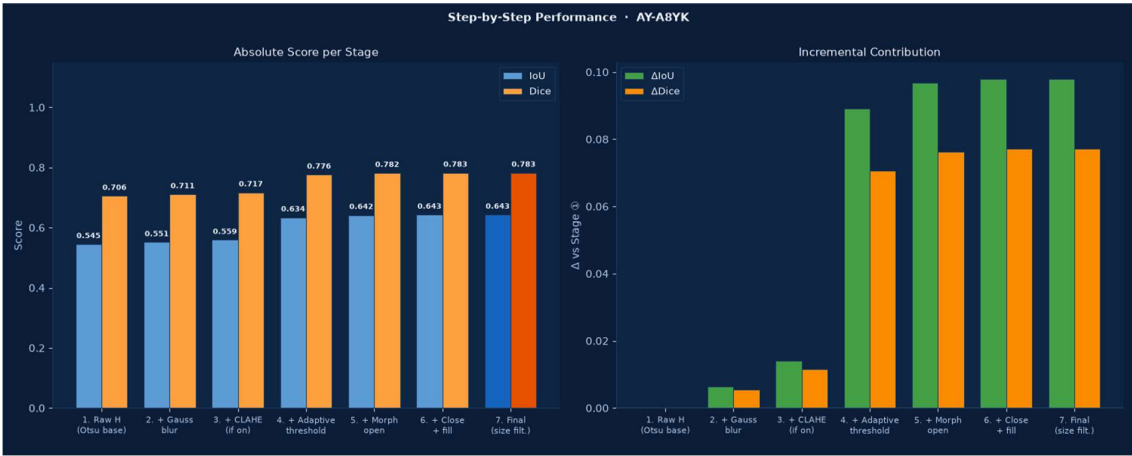


Figure 2.17. Step-by-Step Performance for AY-A8YK Image

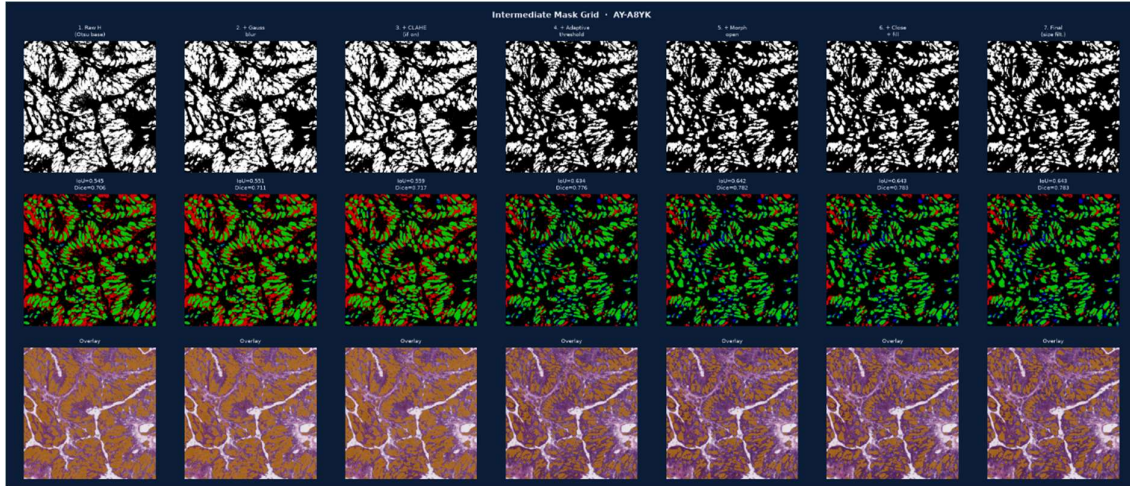


Figure 2.18. Intermediate Mask Grid Result on each Stage for AY-A8YK Image

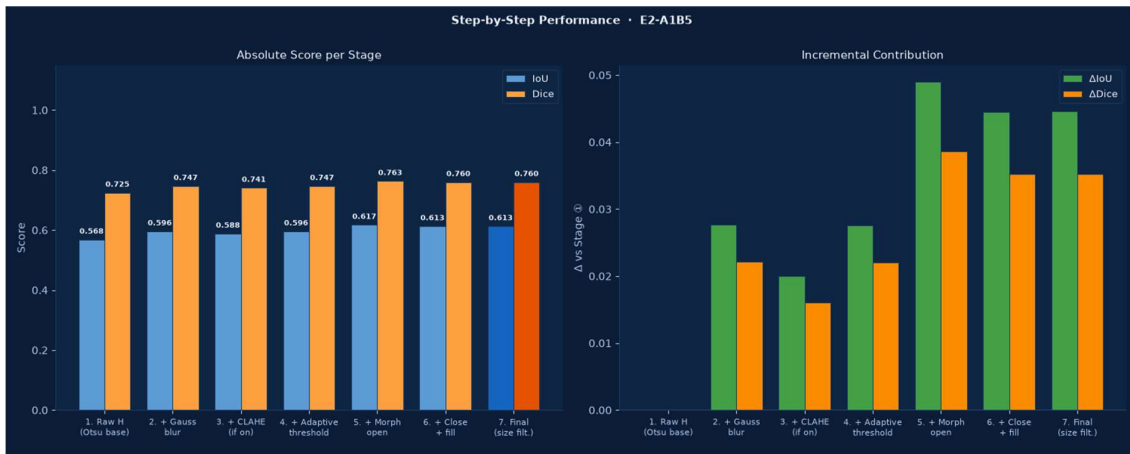


Figure 2.19. Step-by-Step Performance for E2-A1B5 Image

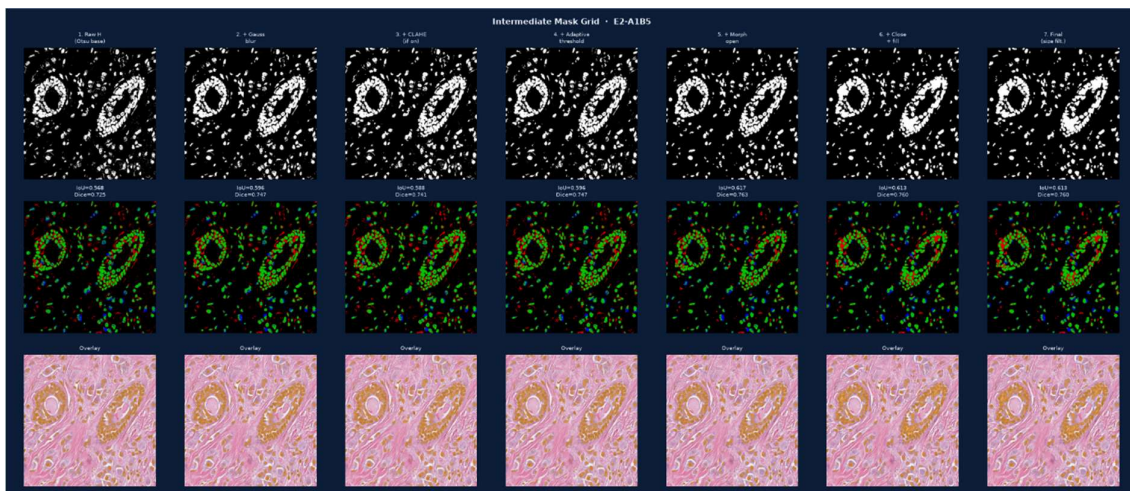


Figure 2.20. Intermediate Mask Grid Result on each Stage for E2-A1B5 Image



Figure 2.21. Step-by-Step Performance for RD-A8N9 Image

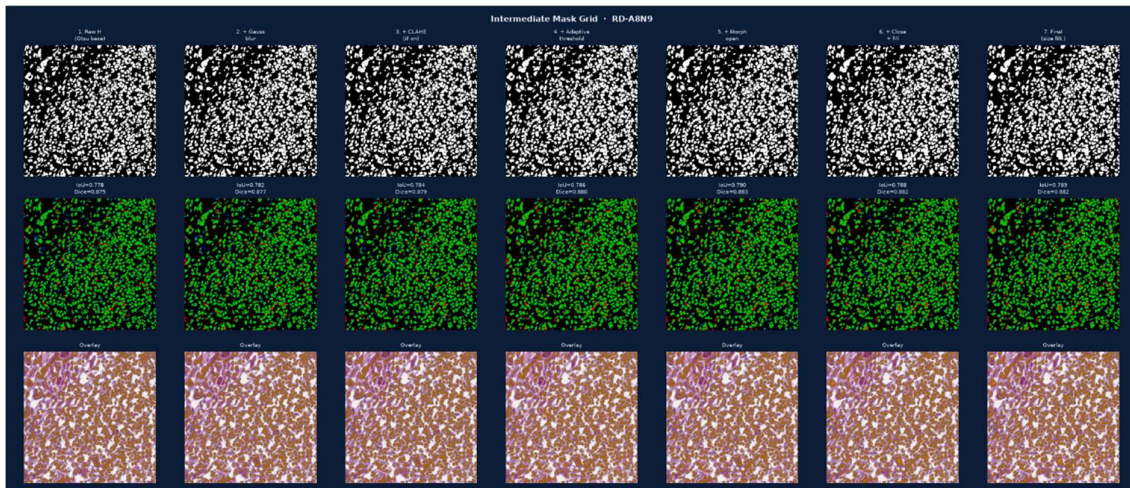


Figure 2.22. Intermediate Mask Grid Result on each Stage for RD-A8N9 Image



Figure 2.23. Comparison of Step-by-Step Performance for All Image

2.3.6. Timing Results

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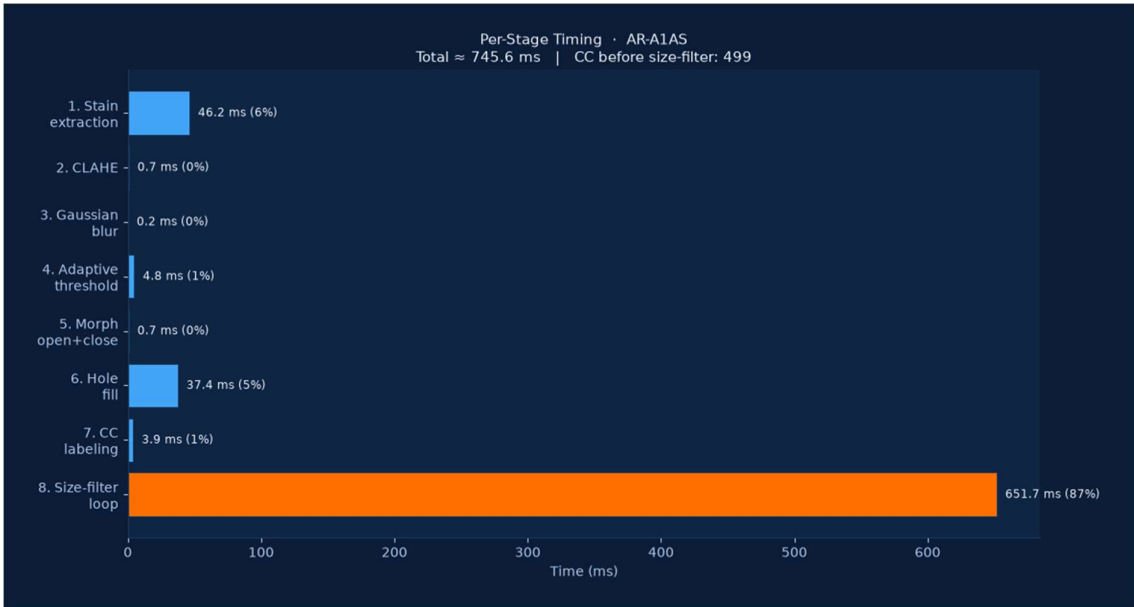


Figure 2.24. Per-Stage Timing for AR-A1AS Image

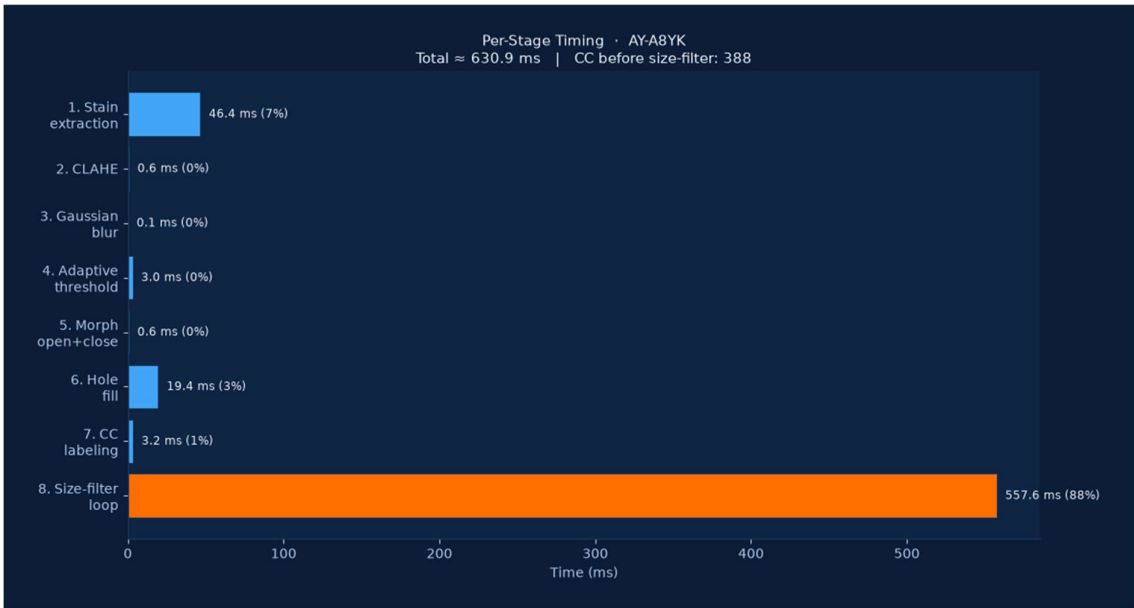


Figure 2.25. Per-Stage Timing for AY-A8YK Image

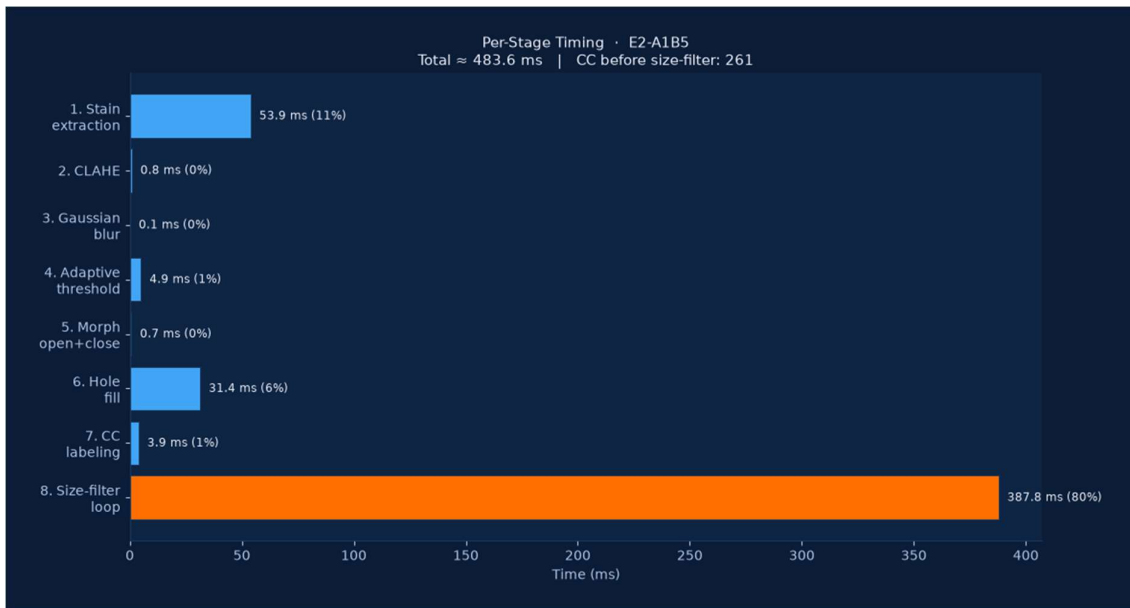


Figure 2.26. Per-Stage Timing for E2-A185 Image

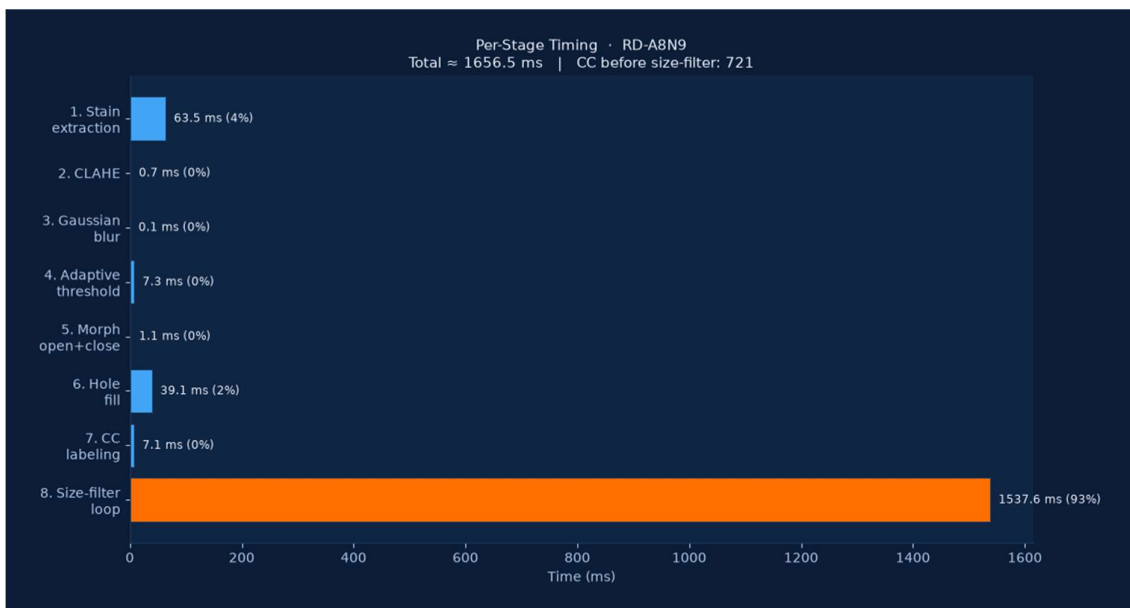


Figure 2.27. Per-Stage Timing for RD-A8N9 Image

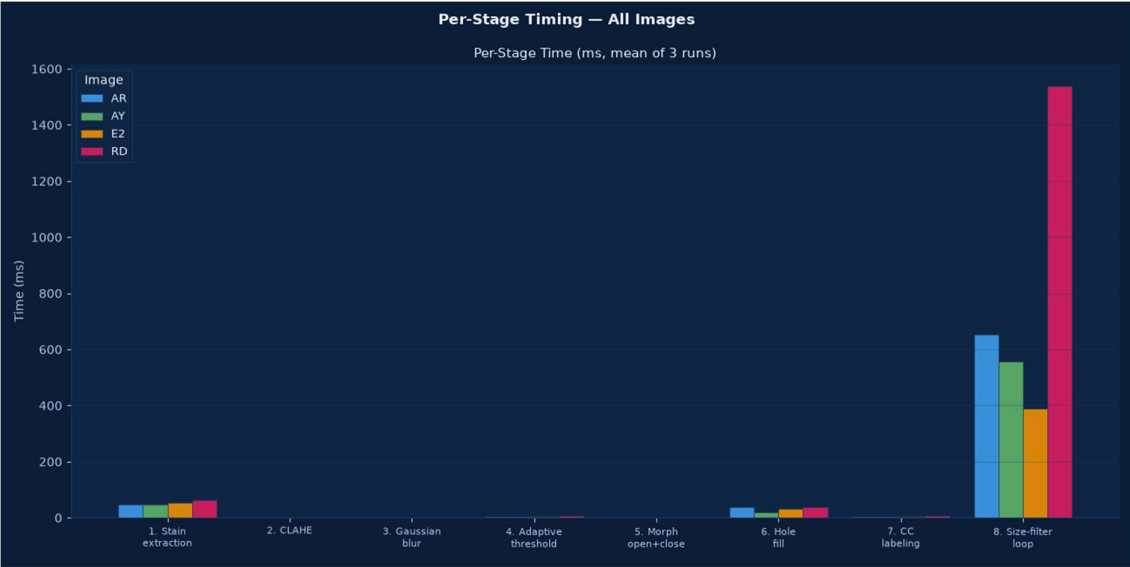


Figure 2.28. Comparison of Per-Stage Timing for All Image

2.3.7. Summary

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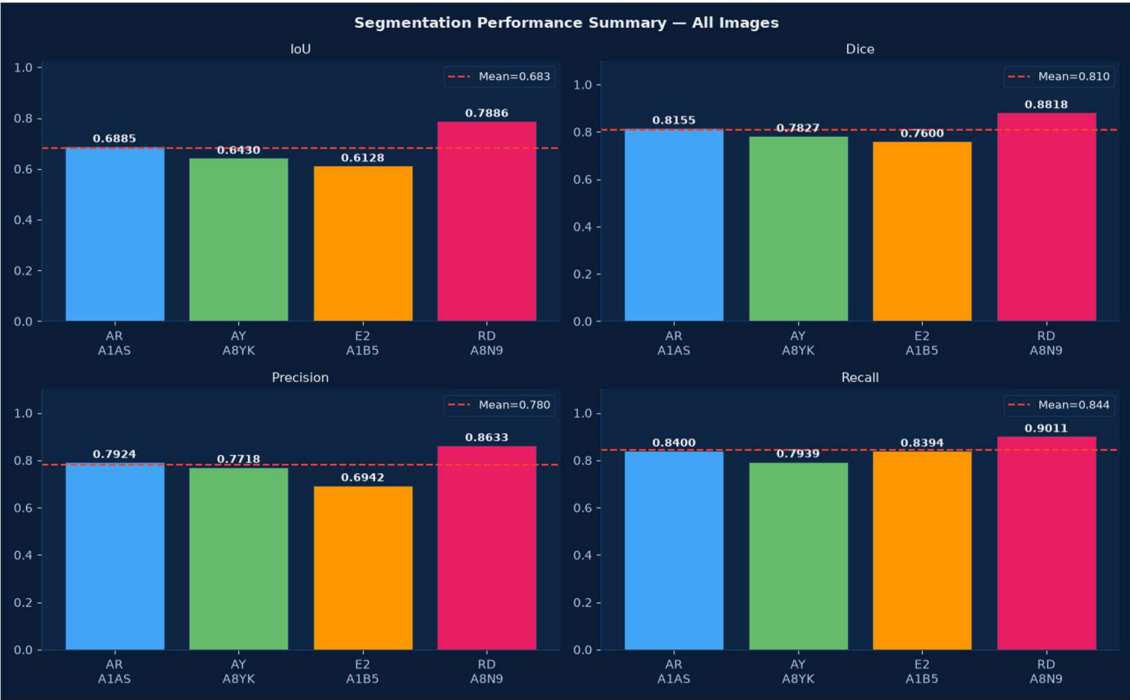


Figure 2.29. Segmentation Performance Summary for All Image



Figure 2.30. Running Time Per Image

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RESULTS SUMMARY — MoNuSeg 2018 · v2.1
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Image	IoU	Dice	Precision	Recall	Running Time	Threshold	Pct
TCGA-AR-A1AS-01Z-00-DX1	0.6885	0.8155	0.7924	0.8400	0.0743	percentile	66.4
TCGA-AY-A8YK-01A-01-TS1	0.6430	0.7827	0.7718	0.7939	0.0749	percentile	68.0
TCGA-E2-A1B5-01Z-00-DX1	0.6128	0.7600	0.6942	0.8394	0.1267	percentile	82.8
TCGA-RD-A8N9-01A-01-TS1	0.7886	0.8818	0.8633	0.9011	0.1077	percentile	61.3

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Mean IoU  : 0.6832 (±0.0768)
Mean Dice : 0.8100 (±0.0530)
Mean Time  : 0.0959 s
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Figure 2.31. Detailed Results Summary in Table-like Format

2.4. Discussion and Evaluation

[discussion and evaluation]

CHAPTER III. CONCLUSION

[General Conclusion]

REFERENCES

- [1] IEEE format